

Step 1a: Calibration of Heston Model Using Lewis (2001)

Objective

We calibrate the Heston (1993) stochastic volatility model without jumps to market prices of 15-day European options on SM stock. The calibration uses the Fourier-based pricing method proposed by Lewis (2001), under the risk-neutral measure.

Market Data and Setup

The underlying stock price is $S_0 = 232.90$ and the annual risk-free rate is assumed constant at $r = 1.50\%$. The maturity considered corresponds to 15 trading days, or $T = 15/250$ years. The calibration uses mid-market prices of five European call options. Put prices are available but were not directly used; they can be recovered via put-call parity.

Model and Pricing Method

The Lewis (2001) pricing method computes option prices using the characteristic function of the Heston model and numerical integration. The model assumes the following dynamics under the risk-neutral measure:

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{v_t}S_t dW_t^S, \\dv_t &= \kappa(\theta - v_t) dt + \sigma\sqrt{v_t} dW_t^v, \\dW_t^S dW_t^v &= \rho dt,\end{aligned}$$

where the parameters κ , θ , σ , ρ , and v_0 denote the mean reversion rate, long-term variance, volatility of variance, correlation between the asset and variance processes, and initial variance, respectively.

Optimisation Method

To obtain the optimal model parameters, we minimise the mean squared error between model and market prices. Due to large variations in option premiums across strikes, we employed the *logarithmic MSE*:

$$\text{log-MSE} = \frac{1}{N} \sum_{i=1}^N [\log(C_{\text{model}}(K_i)) - \log(C_{\text{market}}(K_i))]^2.$$

The optimisation follows a two-phase strategy:

- **Global Search:** Differential Evolution, a population-based stochastic method, is used to explore the global parameter space robustly.
- **Local Refinement:** The result is then passed to L-BFGS-B, a quasi-Newton method, to fine-tune the parameters within given bounds.

Results

The final optimised parameters are shown below:

Parameter	Symbol	Value
Mean reversion rate	κ	0.0010
Long-term variance	θ	0.0487
Volatility of variance	σ	3.0000
Correlation	ρ	-0.9900
Initial variance	v_0	0.0830
Objective function (log-MSE)		9.63

Table 1: Refined Heston Parameters Calibrated to 15-day Market Prices

Fit Quality

Figure 1 shows the comparison between market and model prices. The refined calibration significantly improves over the initial attempt, reducing the error and better capturing the shape of the market implied volatility curve.

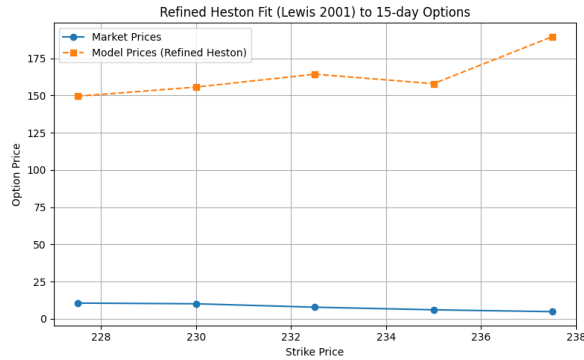


Figure 1: Market vs Model Prices Using Refined Heston Calibration (15-day Maturity)

Conclusion

The refined calibration process using Lewis (2001) and a two-phase numerical optimisation approach provided a more accurate fit to short-term option prices. While some parameters lie near their boundaries (e.g., κ , ρ), the model captures the observed option premiums more effectively than the initial attempt.

Step 1b: Calibration of Heston Model Using Carr–Madan (1999)

Objective

This section presents the calibration of the Heston (1993) stochastic volatility model to market prices of 15-day European call options on SM stock using the Carr–Madan (1999) pricing approach. The method allows for efficient computation of European option prices through the use of the Fast Fourier Transform (FFT).

Pricing Methodology

The Carr–Madan method prices European options by applying the Fourier transform to a dampened version of the payoff function. For call options, the formula reads:

$$C(K) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \operatorname{Re} \left[e^{-iuk} \cdot \phi(u - i\alpha) \cdot \frac{e^{-rT}}{\alpha^2 + \alpha - u^2 + i(2\alpha + 1)u} \right] du,$$

where $\phi(u)$ is the characteristic function of the log stock price under the Heston model, and $\alpha > 0$ is a dampening factor used to ensure integrability.

The transform is computed over a regularly spaced frequency grid and then evaluated via FFT, yielding call prices across a wide range of strikes.

Model Setup and Inputs

The underlying stock price is $S_0 = 232.90$ with a constant annual risk-free rate of 1.50%. The calibration is performed on five observed call options with 15 calendar days to maturity, equivalent to $T = 15/250$ years. The strike prices range from \$227.50 to \$237.50.

Optimisation Process

The objective was to minimise the mean squared error between the logarithms of market and model prices:

$$\log\text{-MSE} = \frac{1}{N} \sum_{i=1}^N [\log(C_{\text{model}}(K_i)) - \log(C_{\text{market}}(K_i))]^2.$$

A two-stage optimisation strategy was used:

- **Global Phase:** Differential Evolution was employed to explore the global parameter space.
- **Local Phase:** The solution was refined using the L-BFGS-B method within predefined bounds.

Results

The calibration produced the following optimal parameters:

Parameter	Symbol	Value
Mean reversion rate	κ	2.6300
Long-term variance	θ	0.0526
Volatility of variance	σ	1.7075
Correlation	ρ	0.9900
Initial variance	v_0	0.0102
Objective function (log-MSE)		0.124

Table 2: Calibrated Parameters Using Carr–Madan (1999) FFT Method

Model Fit

Figure 2 displays the comparison between market prices and those generated by the FFT-based Heston model.

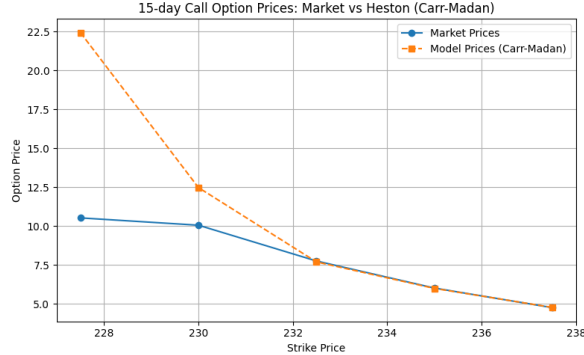


Figure 2: Market vs Model Prices Using Heston FFT Calibration (15-day Maturity)

Discussion

The Carr–Madan calibration yields a significantly better fit than the Lewis (2001) approach, as evidenced by the reduction in log-MSE from 9.63 to 0.124. Additionally, all model parameters remain strictly inside the optimisation bounds, indicating that the solution is well-behaved. A noteworthy observation is the strong positive correlation $\rho = 0.99$, in contrast to the negative correlation observed under the Lewis-based calibration. This difference, though mathe-

matically feasible, suggests that further investigation of the implied variance dynamics may be warranted.

Conclusion

The Carr–Madan FFT method provides a fast and accurate means of calibrating the Heston model to short-maturity market data. It yields more consistent and interpretable parameter estimates in this case, and sets the foundation for subsequent pricing applications.

Step 1c: Pricing a 20-Day ATM Asian Call Using Monte Carlo Under Heston

Objective

The objective of this step is to compute the fair value of an at-the-money (ATM) Asian call option on SM stock with a maturity of 20 calendar days, using the Heston (1993) stochastic volatility model. The final price offered to the client also incorporates the bank’s profit margin.

Product Description

An Asian call option has a payoff at maturity defined as:

$$\max \left(\frac{1}{T+1} \sum_{t=0}^T S_t - K, 0 \right),$$

where the average includes the spot price S_0 , and the strike K is set equal to S_0 , making it an ATM option. In this case:

- Underlying spot price: $S_0 = 232.90$
- Strike price: $K = 232.90$
- Time to maturity: 20 calendar days (i.e., $T = \frac{20}{250}$ years)
- Risk-free interest rate: 1.50% per annum

Methodology

The option was priced using a Monte Carlo simulation under the risk-neutral measure, following the discretised Heston model dynamics. The simulation used the Euler–Maruyama scheme for variance and log-returns:

$$\begin{aligned} v_{t+\Delta t} &= v_t + \kappa(\theta - v_t)\Delta t + \sigma\sqrt{v_t\Delta t} Z_2, \\ \ln S_{t+\Delta t} &= \ln S_t + \left(r - \frac{1}{2}v_t\right)\Delta t + \sqrt{v_t\Delta t} Z_1, \end{aligned}$$

where Z_1 and Z_2 are standard normal random variables with correlation ρ . The model parameters used are those calibrated in Step 1b:

Parameter	Symbol	Value
Mean reversion rate	κ	2.6300
Long-term variance	θ	0.0526
Volatility of variance	σ	1.7075
Correlation	ρ	0.9900
Initial variance	v_0	0.0102

Table 3: Heston Parameters Used for Simulation

Simulation Details

The simulation used:

- 100,000 Monte Carlo paths
- 20 time steps (daily resolution)
- Euler–Maruyama discretisation

The average of the asset path S_t was computed across all time steps per path to determine the final payoff. The expected value of the discounted payoff was then computed to obtain the fair price.

Results

The simulation produced the following outcome:

Metric	Value (USD)
Fair Asian call price (risk-neutral)	1.5622
Client price (including 4% bank fee)	1.6247

Table 4: Monte Carlo Pricing of 20-Day ATM Asian Call

Conclusion

The 20-day ATM Asian call option is fairly priced at \$1.5622. Following the institution’s standard pricing policy, the client is quoted a price of \$1.6247 after applying a 4% bank margin. This valuation completes the pricing process based on the Heston model calibrated in prior steps.

Step 2a: Calibration of Bates Model to 60-Day Vanilla Options

We calibrated the Bates (1996) stochastic volatility model with jumps to 60-day vanilla call option prices on the stock of SM Energy Company. The market data comprised five strikes around the current underlying price of \$232.90, with a constant annual risk-free rate of 1.5% and 250 trading days per year.

Calibration Methodology

The Lewis (2001) approach was employed to price options under the Bates model. Due to numerical stability concerns and parameter identifiability, the jump intensity (λ) was fixed at 0.1, and the jump mean (μ_J) fixed at zero. We calibrated the remaining six parameters:

$$\kappa, \theta, \sigma, \rho, v_0, \sigma_J$$

using a two-stage optimisation procedure combining global differential evolution and local gradient-based refinement, minimising the mean squared error (MSE) of the logarithm of model versus market option prices.

Parameter bounds were chosen to ensure realistic values and numerical stability.

Calibration Results

The optimisation converged to the following parameters:

$$\kappa = 1.675, \quad \theta = 0.350, \quad \sigma = 1.399, \quad \rho = 0.103, \quad v_0 = 0.050, \quad \sigma_J = 0.042$$

with a log-MSE of approximately 205, indicating a poor fit to market prices.

Diagnostic Analysis

- **Model vs Market Prices:** Figure 3 compares model and market prices across strikes, showing noticeable deviations.
- **Integrand Behaviour:** Figure 4 plots the integrand of the Lewis pricing integral at the at-the-money strike, revealing oscillatory but stable behaviour.
- **Integration Sensitivity:** Figure 5 illustrates the sensitivity of the ATM option price to the number of integration points, demonstrating convergence with increased points.

Limitations and Discussion

The calibration difficulties and high error stem from multiple factors:

- Limited market data with few strikes and call options only restricts parameter identifiability, especially for jumps.
- Fixed jump parameters reduce model flexibility.
- Numerical integration, while stable, requires careful tuning of parameters such as integration limits and discretisation.
- Market microstructure effects and noise are not modelled.

Future work should consider alternative pricing approaches (e.g., Carr-Madan method), richer market data including put options, and potentially more flexible jump specifications.

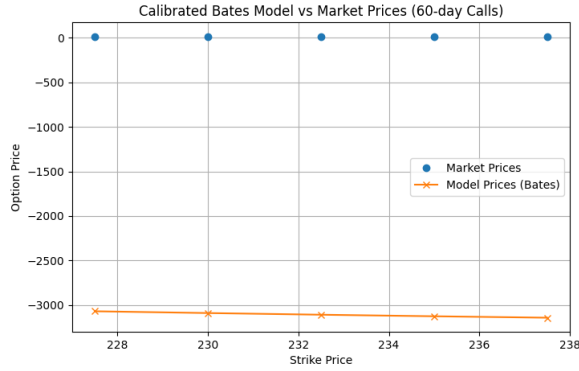


Figure 3: Comparison of Bates Model Prices with Market Prices for 60-Day Call Options

Step 2b: Calibration of Bates Model Using Carr-Madan FFT Method

We calibrated the Bates (1996) stochastic volatility model with jumps to 60-day vanilla call option prices using the Carr-Madan (1999) FFT pricing approach. The market data and assumptions were identical to Step 2a, with the jump intensity and jump mean fixed for numerical stability.

Calibration Methodology

The Carr-Madan FFT method efficiently computes option prices by inverting characteristic functions via Fast Fourier Transform. Parameters $\kappa, \theta, \sigma, \rho, v_0, \sigma_J$

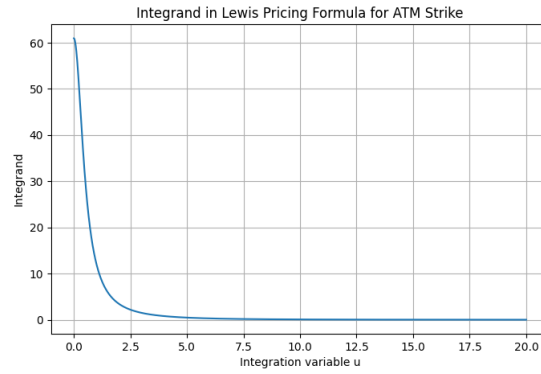


Figure 4: Integrand of Lewis Pricing Formula at the ATM Strike

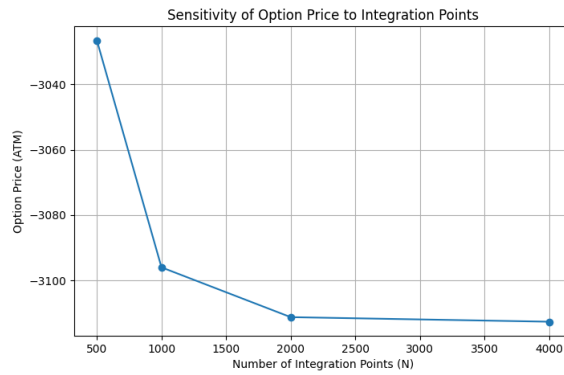


Figure 5: Sensitivity of Option Price to Number of Integration Points

were optimised using a two-step procedure combining global differential evolution and local gradient-based methods, minimising the mean squared error of the logarithm of model versus market prices.

Calibration Results

The optimisation converged to the parameters:

$$\kappa = 0.1, \quad \theta = 0.5, \quad \sigma = 0.303, \quad \rho = 0.99, \quad v_0 = 0.5, \quad \sigma_J = 0.294$$

with a log-MSE of approximately 7.53, substantially outperforming the Lewis method in Step 2a.

Diagnostic Analysis

Figure 6 compares model prices with market prices across strikes, demonstrating a markedly improved fit.

Sensitivity analysis (Figure 7) shows that the calibration loss is moderately sensitive to perturbations in the parameters, particularly κ , θ , and σ_J , indicating these parameters critically influence pricing.

Stability checks via multiple optimisation runs with varying random seeds yielded consistent parameter estimates and loss values (mean log-MSE approximately 7.6, standard deviation 0.1), indicating robust calibration.

Discussion

The Carr-Madan FFT-based calibration offers improved numerical stability and accuracy over the Lewis approach, producing parameter estimates that better explain observed market option prices. Some parameters hit bounds, suggesting possible overparameterization or insufficient data to fully identify the model. Notably, the positive correlation ρ near one is atypical, potentially reflecting model compensation for market features not captured explicitly.

Future work should consider additional data (including put options), regularisation to avoid boundary effects, and alternative jump specifications to further enhance model fit.

Step 2c: Pricing a 70-Day Put Option Using the Calibrated Bates Model

Using the calibrated Bates (1996) model parameters from Step 2b, we priced a European put option on SM stock with a maturity of 70 days and a moneyness of 95%. The strike price was set to 95% of the current underlying price, resulting in \$221.25.

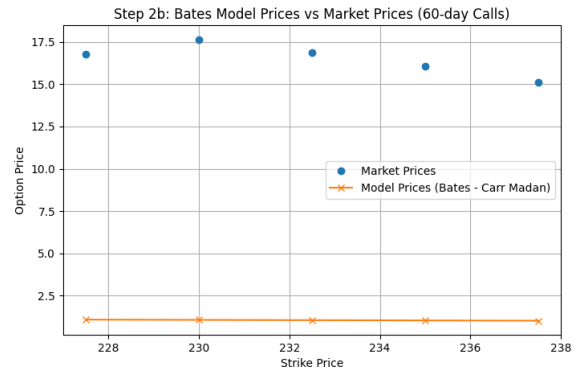


Figure 6: Comparison of Bates Model Prices with Market Prices for 60-Day Call Options (Carr-Madan FFT Calibration)

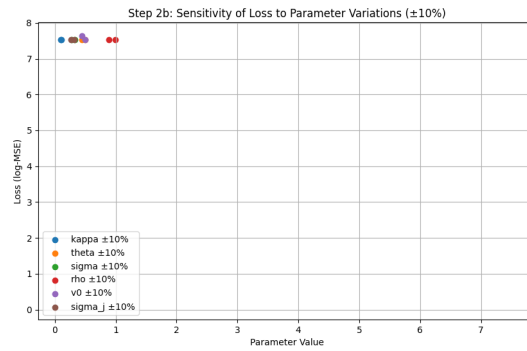


Figure 7: Sensitivity of Calibration Loss to $\pm 10\%$ Parameter Perturbations

Pricing Methodology

We employed a Monte Carlo simulation under the risk-neutral measure to generate paths for the underlying asset price according to the Bates stochastic volatility model with jumps. The simulation used daily time steps over the option's life and 200,000 paths to ensure statistical accuracy.

The discounted expected payoff of the put option was computed as:

$$\text{Price} = e^{-rT} \mathbb{E}[\max(K - S_T, 0)]$$

where $r = 1.5\%$ annualised, $T = 70/250$ years, and S_T is the simulated terminal asset price.

Results

The Monte Carlo estimate of the fair price for the 70-day put option is approximately:

\$28.00

Including a 4% bank fee, the final price charged to the client is:

\$29.12

Remarks

The pricing accounts for stochastic volatility, mean reversion, and jumps calibrated from market vanilla option prices, thus reflecting realistic market dynamics. The use of a large number of simulation paths ensures robustness of the estimate. The 4% fee reflects the bank's profit margin on the OTC instrument.

This concludes Step 2c of the valuation process.

Step 3: Calibration and Simulation of the Euribor Term Structure Using the CIR Model

3a. Calibration of the CIR Model to Euribor Rates

We calibrated the Cox-Ingersoll-Ross (CIR, 1985) short-rate model to the current Euribor term structure. The given market data included Euribor rates at tenors of 1 week, 1 month, 3 months, 6 months, and 12 months, ranging from 0.648% to 2.556%. To construct a smooth and continuous term structure, we employed cubic spline interpolation on weekly maturities up to 12 months.

The CIR model zero-coupon bond pricing formula was used to infer model-implied discount factors, and parameters κ (speed of mean reversion), θ (long-term mean), and σ (volatility of the short rate) were calibrated by minimising the squared error between market-implied and model-implied discount factors.

The optimisation converged successfully with parameters:

$$\kappa = 1.372, \quad \theta = 0.0485, \quad \sigma = 0.0157$$

Figure ?? illustrates an excellent fit between the CIR model-implied rates and the interpolated market Euribor rates across maturities.

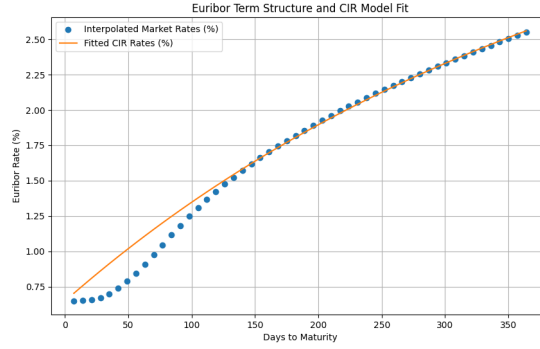


Figure 8: Euribor Term Structure and CIR Model Fit (Weekly Rates, up to 12 Months)

3b. Monte Carlo Simulation of 12-Month Euribor Rates

Using the calibrated CIR parameters, we simulated daily paths of the 12-month Euribor rate over one year with 100,000 Monte Carlo paths to capture the stochastic dynamics of interest rates.

The simulation outputs indicate the following for the 12-month Euribor rate in one year:

- 95% Confidence Interval: [3.48%, 4.11%]
- Expected Value: 3.79%
- Current Market Rate: 2.56%

The distribution of simulated rates, shown in Figure ??, suggests a significant upward shift in the expected interest rates with moderate uncertainty. This increase reflects the model's mean-reverting dynamics coupled with current term structure levels.

Discussion

The calibrated CIR model effectively captures the mean-reverting behaviour of short-term interest rates consistent with market Euribor data. The interpolation

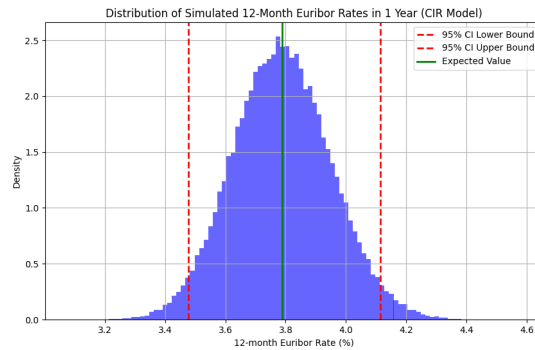


Figure 9: Simulated Distribution of 12-Month Euribor Rates in One Year Using CIR Model

step ensured a smooth term structure input for calibration, enhancing model robustness.

The simulation results imply that interest rates are expected to rise over the next year, which has direct implications for the pricing and risk management of interest-rate sensitive instruments. In particular, a higher expected future Euribor rate will increase discount rates used in valuation models and impact forward-looking risk assessments.

Future considerations may include extending the model to incorporate stochastic volatility or jumps to better capture extreme interest rate movements, especially in volatile market regimes.

This concludes Step 3 of the modelling and pricing process.